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Synthetic Relativity versus Situation Calculus versus GLB

Aspect	*Synthetic Relativity (R-DTLF)*	*Situation Calculus (BATs/SSAs) $+$ variants of μ-calculus*	*Bimodal Provability logic (GLB)*
Modalities	RDo, ODo & PDo	PDo, $\langle a\rangle\phi$ (possibility), $[a]\phi$ (necessity)	$\Box$, $\Box_\omega$
Ordering	Sequences of finite Posets, non-unique predecessors, non-well ordered	Successor function, Unique predecessors, non-well ordered	Transfinite ordinality: limits, Non-unique predecessors, well-ordered
Completeness?	Spatially complete (continuous geometry) and Turing complete (universal computation via recursive domains); Arithmetically incomplete (triggers Gödel's incompleteness).	Turing complete (can compute anything), but neither spatially complete (no continuous geometry) nor arithmetically complete (subject to Gödel's incompleteness).	Arithmetically complete (flawlessly models the bounds of provability), but neither Turing complete (strictly decidable, cannot compute loops) nor spatially complete.
Other	Algebraic compactness. Kleene's Realizability & BHK (Brouwer-Heyting-Kolmogorov) Interpretation.	Regression Theorem (solves the frame problem). Fixpoint Semantics & Knaster-Tarski Theorem.	Japaridze's Arithmetical Completeness Theorem. Löb's Theorem & Well-founded Kripke Semantics.

- RDo is equivalent to the $\Box$ modality of bimodal provability logic GLB: *"There exists a finite, step-by-step mechanical proof of this property."*
- ODo is equivalent to the $\Box_\omega$ modality of GLB: *"Evaluated at the continuous limit, this property is universally true,"* i.e., this is the modality of *semantic reflection*.

Theory-Driven Generalization

When the converged limit shifts to become the new Reference Frame (S_f) via the **Transformation Axiom**, it acts natively as a global Greatest Fixpoint (ν) and the coinduced property is added to the reference theory for subsequent computations. The system treats **generalisation** as a categorical operation over its Reference Theory (S_F) facilitating invariant proposal through the foundational duality of fibrations and adjunctions.

- **1. Change of Base (The Fibrational Shift)**
  - Logical properties form a *Total Category* fibered over a *Base Category* of states.
  - The system performs a **Change of Base** from the coinduced space (ODo) to the inductive space (RDo). This functor maps complex, induced histories into coinduced progressions.
- **2. Adjoint Duality (Swapping the Problem)**
  - Generalising a safe invariant is structurally identical to the problem of deductive planning, simply with the functors reversed (an Adjunction: $Left \dashv Right$).
  - *Planning (Left Adjoint):* Inductive, bottom-up synthesis of a finite trace (Least Fixpoint).
  - *Generalising (Right Adjoint):* Coinductive, top-down projection of an infinite bound (Greatest Fixpoint). The finite local observations mathematically push across the adjunction to form the exact coinductive proposal.

Theory-Driven Generalization (continued)

- **3. Universal Properties (Terminal Coalgebras)**
  - The system requires no external engineering templates or “Syntax-Guided” rules.
  - Because S_F natively defines the geometric symmetries of the domain, the coinductive proposal emerges as a **Universal Property** (specifically, a Terminal Coalgebra). Stone Duality ensures that the logical syntax automatically folds to perfectly bound the state space’s innate topology.

Domain Theory in Logical Form (DTLF)

Introduced by Samson Abramsky (1991), DTLF provides the foundational “physics” of computation. To map the true topology of reasoning, we must first abandon classical, absolute logic.

- **The Classical Flaw:** Standard logic (Boolean) assumes an absolute, “God’s eye” view of reality (True/False). It relies on the Law of Excluded Middle ($A \vee \neg A$) and ignores *how* a system physically computes truth.
- **Intuitionistic & Constructive:** DTLF operates from the inside out. Truth is not a static given; it must be actively constructed step-by-step.
- **The Logic of Partial Information:** In DTLF, variables do not represent absolute boolean states; they represent *observable properties* and evolving states of partial knowledge.

The Scott Domain

In DTLF, computation is an upward, continuous climb from baseline ignorance toward a fully realised concept within a geometric **Scott Domain**.

- **Baseline Ignorance ($\perp$):** The logic does not start at "False." It starts at Bottom ($\perp$)—a formal mathematical coordinate representing zero information and zero computation.
- **Compact Elements (The Primitives):** The finite, discrete pieces of verifiable evidence a system can physically observe (e.g., *a specific sensor reading*, or *a sequential token*).
- **Directed Accumulation:** Knowledge is actively gathered. The system continuously accumulates these compact elements, generating a directed proof path ($\sqsubseteq$).

Spatial Completeness

The profound mathematical power of vanilla DTLF lies in Abramsky's **Spatial Completeness Theorem**, rooted in Stone Duality.

It proves a mathematically perfect, 1-to-1 isomorphism between:

1. **The Syntax:** The algebraic, step-by-step logical code ($\phi \vdash \psi$).
2. **The Semantics:** The continuous, geometric space of concepts ($[\![\phi]\!] \subseteq [\![\psi]\!]$).

The Guarantee: Every step of logical deduction perfectly corresponds to navigating a continuous topology. Computations resolve exactly when the directed path of evidence catches a continuous topological limit (Supremum, $\bigsqcup$).

Limitation: Vanilla DTLF perfectly describes a single, static universe. To model continuous learning and dynamic abstraction, the Reference Frame

The Core Philosophy

Exogenous Truth vs. Endogenous Observation

- **Classical Logic is Exogenous:** It looks at abstract, eternal truths from the outside.
- **Vanilla DTLF is Endogenous:** It is built from the inside out.
- **Observable Properties:** In DTLF, a formula ϕ does not represent an absolute "Truth." It represents a finite, computationally *observable property* (a compact open set in the Scott Topology).
- **The Act of Verification:** To assert ϕ is to mathematically state: *"Based on a finite amount of inductive computational steps, I have actively gathered the evidence to verify property ϕ."*

The Formal Syntax

A Strictly Positive Propositional Logic

Because a system can only assert what it can finitely verify, the baseline syntax is strictly positive:

$$\phi, \psi ::= \mathbf{t} \mid \mathbf{f} \mid \phi \wedge \psi \mid \phi \vee \psi \mid \mathrm{Constructors}(\phi)$$

- **t (True / Baseline):** The empty observation. Requires zero information (topologically, the bottom $\perp$ of the domain, where you know nothing).
- **f (False / Contradiction):** Impossible information (a computational crash).
- $\phi \wedge \psi$ **(Conjunction):** The active accumulation of evidence (both verified).

The Engine & Catching Limits

Entailment ($\vdash$) and Syntactic Fixpoints

Instead of static Boolean truth tables, DTLF uses an **Entailment Relation** to mathematically move "up" the directed poset towards a limit. $\phi \vdash \psi$: *"Any state of information satisfying ϕ automatically contains enough information to satisfy ψ."* - **Rules:** Reflexivity ($\phi \vdash \phi$), Transitivity (if $\phi \vdash \psi$ and $\psi \vdash \chi$, then $\phi \vdash \chi$), Conjunction ($\phi \land \psi \vdash \phi$).

How Vanilla DTLF Catches Limits: Because syntax is finite, an infinite continuous limit cannot be a single finite formula. - **Limits as Filters:** A semantic limit is a *Filter* or *Ideal*—an infinite, directed set of formulas mutually entailing each other ($\phi_1 \vdash \phi_2 \vdash \phi_3 \ldots$). - **The Fixpoint Operator (μ):** To handle recursive loops, vanilla DTLF relies on a static, syntactic variable axiom ($\mu X.\, F(X)$) to declare that a least fixpoint exists.

Categorical (Stone) duality: Domain Theory in Logical Form (DTLF) & Scott topologies

Through Stone duality, DTLF proves that these two sides are mathematically isomorphic:

- The Spatial Side (Semantics): Domains of computational values equipped with the Scott topology.
- The Algebraic Side (Syntax): The logic of observable properties, structured as an algebraic lattice.

To capture the continuous reality of the logical observer, we must transition to the semantics of **DTLF**.

Abramsky, S. (1987). *Domain theory in logical form*. Proceedings of the Symposium on Logic in Computer Science (LICS '87), 47-53. IEEE.

Abramsky, S. (1991). *Domain theory in logical form*. Annals of Pure and

DTLF spaces are T_0 (Kolmogorov) spaces (Why it is non Hausdorff)

For a topological space to be Hausdorff (T_2), you must be able to take any two distinct points x and y and put them in completely separate, non-overlapping “bubbles” (disjoint open sets).

In domain theory, points are not independent locations in space; they represent stages of computation ordered by information content ($x \sqsubseteq y$ means “x is a partial approximation of y”).

Because open sets in DTLF represent observable properties, the Scott topology dictates that every open set must be upward-closed. If an observable property is true of a partial computation x (e.g., “it has printed the letter A”), it must remain true for any further refinement y (e.g., “it has printed A, B, and C”). Information is monotonic; it cannot be “un-learned.”

Compactness & finite observability

The syntax (the logic): Bounded Distributive Lattices

The semantics (the geometry): Spectral spaces

Compact open sets & finite observability:

Compactness bridges infinite mathematical semantics with finite computational syntax.

Qualities of DTLF

The Scott Topology perfectly matches a formal, inductive logic: Open sets in the Scott Topology The Logic (Syntax): A formula in standard DTLF represents a finite, verifiable piece of information. The Space (Semantics): This formula maps perfectly to a compact open set (a finite approximation) in the Scott Topology of the domain. The Proof Paths: Limit creation in the domain (taking the supremum of a directed set) corresponds exactly to logical entailment.

- Nets indexed by directed sets.
- Compact elements

A compact element (sometimes called a finite element). An element c is compact if, whenever c is less than or equal to the limit of a directed set, c is actually less than or equal to some element already inside that directed set.

Sequences of Finite Posets (SFPs): Partial ordering

Sequences of Finite Posets (SFP), often called Bifinite domains, were introduced by Gordon Plotkin in 1976, they are domains constructed entirely as the limit of finite approximations. They are the 'gold standard' in DTLF because they are robust enough to model complex features like higher-order functions and non-determinism, while retaining the exact topological properties—compactness—required for finite observability and Stone Duality

- In Computation (Semantics): It models the growth of information ($x \sqsubseteq y$ means x approximates y).
- In Topology (Scott Topology): It dictates what is observable (properties must be "upward-closed" along the order).
- In Logic (Syntax): It defines proof rules and deduction ($A \leq B$ means A entails B).

Comparison with Inductive Logic Programming (ILP)

To prove A relative to B with respect to C is to reject the notion of an absolute, context-free proof. In this ternary relation, C (S_r) serves as the topological Reference Frame (the environment or inductive bias). A (RDo) is the active, syntactic accumulation of finite evidence. B (ODo) is the evaluated semantic limit.

Therefore, the proof formally guarantees that within the strict geometric bounds of frame C, the evidence A is structurally sufficient to synthesise the invariant meaning B.

- Unlike classical Inductive Logic Programming (ILP) where background knowledge is static and human-coded, or standard ML where inductive bias is fixed globally by architecture, the Synthetic Relativity framework is dynamic.
- The system's background knowledge (Frame C) continuously evolves via the Transitivity Axiom, allowing the network to

Comparison with Inductive Logic Programming (ILP) (continued)

Gordon Plotkin proved in 1971 that the Relative Least General Generalisation (RLGG) of two clauses—relative to a background theory of arbitrary Horn clauses—may not exist as a finite clause under θ-subsumption.

- Inverse entailment: grounding background knowledge relative to a specific example, IE constructs the most specific finite clause that entails the example. The search space for the algorithm is then strictly bounded between the empty clause (the most general) and this finite bottom clause (the most specific), completely circumventing the need to generate an infinite RLGG (Muggleton, 1995).
- Extracting structure from higher order ILP (Cropper et al. 2020))